FUTURE OF WORK · GOOGLE AI BUILDER CUP

ANVIL

Governance for autonomous spending agents.

A human signs what an agent may spend. Anvil checks every action against it, blocks drift, and makes the agent fix itself.

THE PROBLEM

Agents already spend money. Nothing checks they stayed inside what a human approved.

- Travel & expense agents book flights and hotels on their own today.
- Rules engines only see numbers: a price under the cap passes, even when the **cabin class**, **refundability** or **intent** is wrong.
- When something goes wrong, there is no tamper-evident record of who authorized what.

Example: "within budget" but out of policy

Agent books **business class** on a non-preferred airline, $590 against a $600 cap. Every budget rule passes. The manager never authorized it.

[ADD: one sourced statistic from the project brief on agentic spend / T&E leakage, with source]

WHAT ANVIL IS

A signed mandate in, a verdict and an audit trail out.

1 · Signed mandate
Manager's limits, signed (HMAC) plus AP2 open mandates signed with the manager's own key

→

2 · Agents over A2A
Flight, hotel, transport agents discover suppliers via Agent Cards, JSON-RPC

→

3 · Anvil auditor
Gemini judges the action against the intent of the mandate; structured JSON

→

4 · Decide
MATCH allow + Cart Mandate
REVIEW one-tap re-consent
DRIFT block + explain

→

5 · Provenance
Hash-chained log with delegation chain, live on the dashboard

Runs on Cloud Run (orchestrator + supplier agents), Gemini on Vertex AI, Firestore, Firebase Hosting.

THE TWIST

Anvil doesn't just say no. The agent corrects itself.

BLOCKED

Emirates · business · $590

"Flight class 'business' does not match mandated 'economy' class."

→

SELF-CORRECTED · MATCH

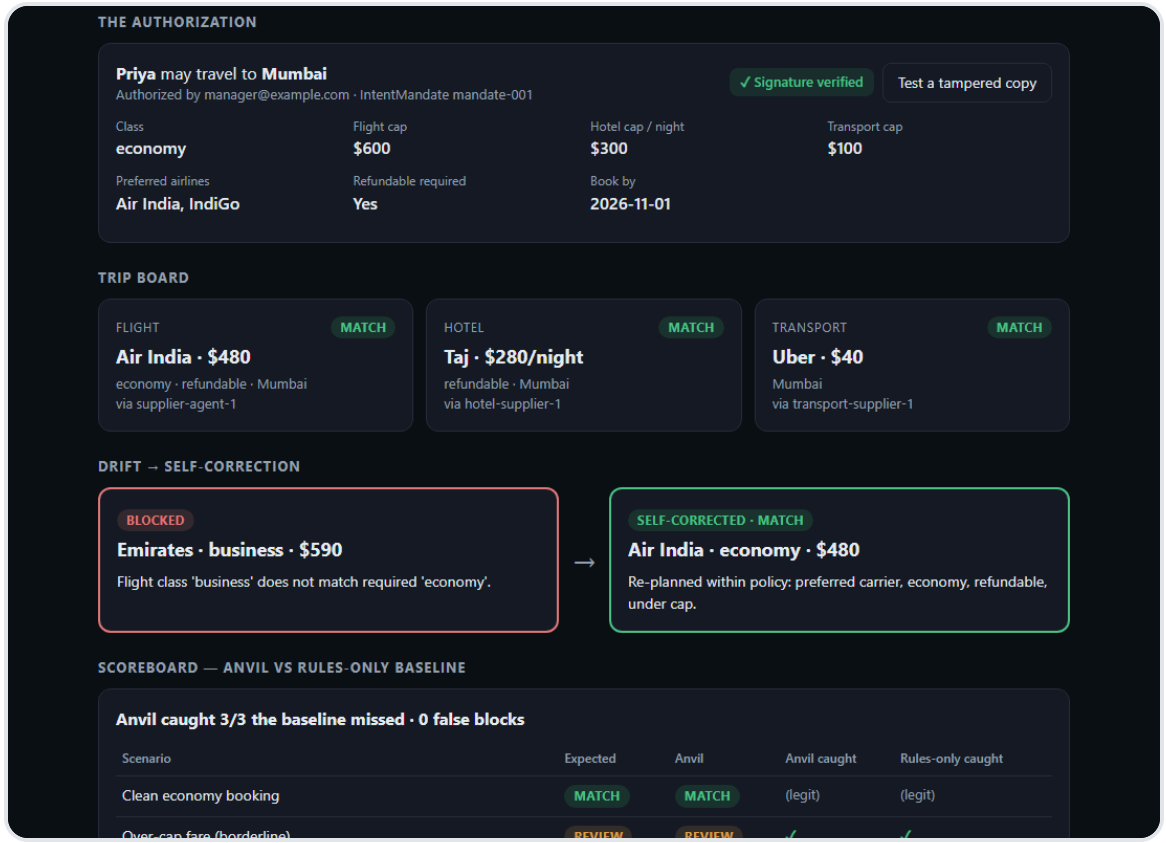
Air India · economy · $480

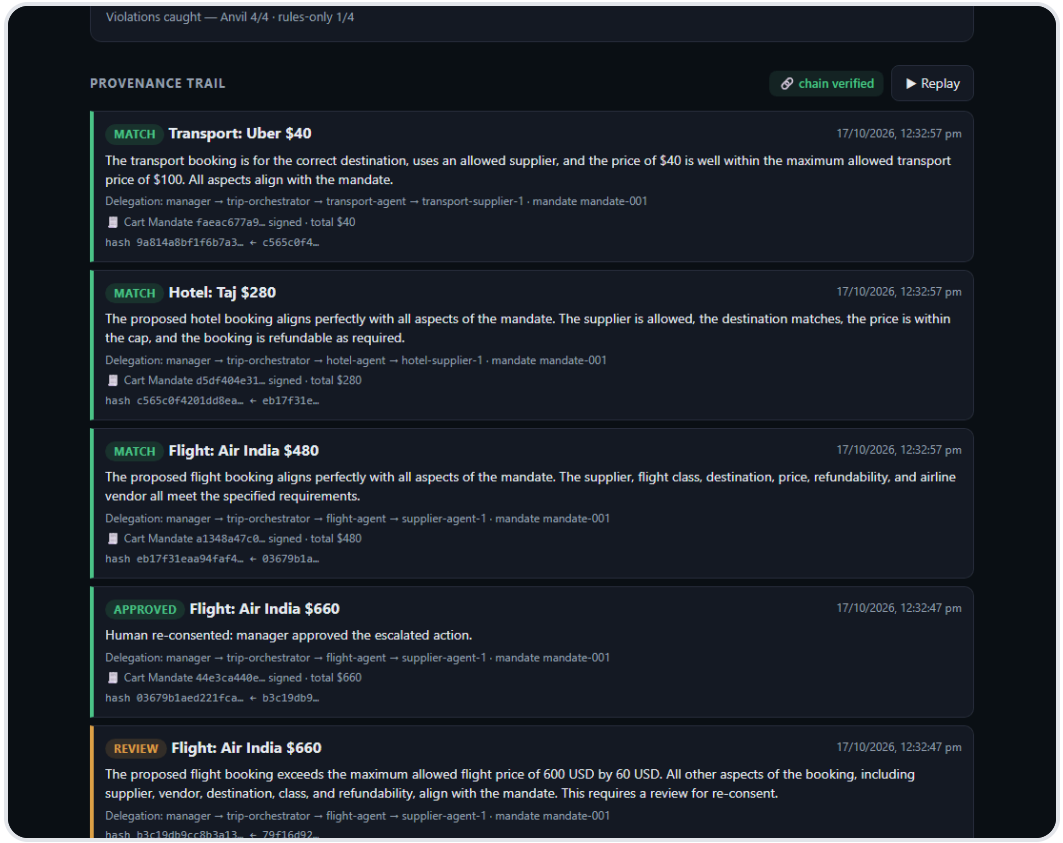
Re-planned within policy: preferred carrier, economy, refundable, under cap.

- Intent-drift reasoning: cabin class, refundability, and a booking **split into two charges to dodge an approval cap**.
- Borderline cases are REVIEW, not blocks: escalated for one-tap manager approval.

LIVE DEMO

One mandate. Three agents. One trail.





Live on Google Cloud. Left: signed mandate, trip board, drift → self-correction. Right: hash-chained trail with delegation and Cart Mandates.

THE PROOF

Measured, not asserted: same actions, two systems.

FAILURE TYPE	ANVIL	RULES-ONLY
Overage (price above cap)	3/3	3/3
Scope (wrong destination, supplier)	3/3	3/3
Substitution (class, refundability)	3/3	0/3
Structuring (split to dodge a cap)	2/2	0/2
Smuggling (hidden add-ons)	2/2	0/2
Injection (fake “pre-approved” notes)	6/6	4/6

19/19

violations caught by Anvil

10/19

caught by a rules-only baseline

0

false blocks on legitimate actions

26 scenarios (7 legitimate, 19 violations incl. 6 adversarial) × 3 live-Gemini repeats: 26/26 stable, 26/26 matched their label, 0 from cache. Reproduce: `python -m anvil.eval` or `/api/eval`.

GOOGLE STACK

Built end to end on Google's agent stack.

Gemini · Vertex AI

The drift auditor: structured JSON verdicts at temperature 0, with retries and a cached-verdict fallback.

A2A

Agent Cards + discovery; Message/Task over JSON-RPC between buyer and supplier agents.

AP2

Official AP2 V2 SDK: every approved booking carries a verified SD-JWT mandate chain; its constraints are a second, deterministic gate.

Cloud Run

Two services: orchestrator + guard + API, and the A2A supplier agents.

Firestore

Mandates and the hash-chained, tamper-evident provenance log.

Firebase Hosting

Live dashboard with a real-time Firestore listener.

Supplier agents pass Google's official A2A client (3/3); mandate chains verified by Google's official AP2 SDK. Buyer client is hand-written; not claimed fully compliant.

HOW IT BROADENS

Beachhead: travel & expense. Destination: the control plane for agent spending.

Today · T&E

Flights, hotels, ground transport governed by one signed mandate.

Next · Cloud & SaaS

Agents that provision infrastructure and buy seats, checked against the same kind of mandate.

Then · Procurement

Vendor selection and purchase orders, with delegation chains and approvals built in.

Same engine, same three verdicts, same tamper-evident trail. Only the mandate and the supplier agents change.

WHY NOW · IMPACT

Agents are live today, and spend leaks without a guard.

[ADD]

Agentic travel/spend already in production (name the real products + source from the brief)

[ADD]

Size of T&E leakage / out-of-policy spend (figure + source from the brief)

[ADD]

Time saved by one-tap re-consent vs manual review (only if measured)

Anvil gives finance and compliance teams what they will need before they let agents spend: proof of authorization, a block on drift, and a trail an auditor can verify.

TEAM · ASK

Let agents spend, with a human's limits intact.

Team

[ADD: team name, members and roles]

Ask

[ADD: what you want from the judges / next step]

Live: https://dazzling-lodge-511607-m0.web.app
Code: https://github.com/Adhi1402/anvil